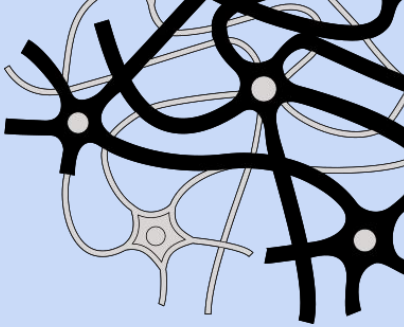
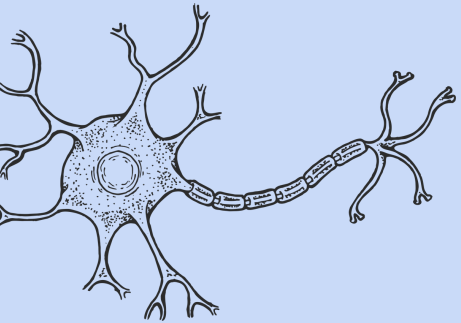




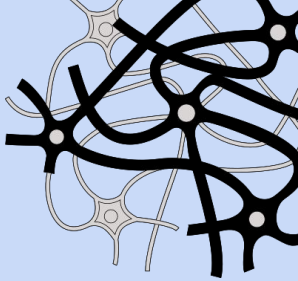
Detekcija intracerebralnog krvarenja na CT slikama mozga

Karla Budimir
Anja Obuljen
Marko Sršić
Nela Štubelj









UVOD

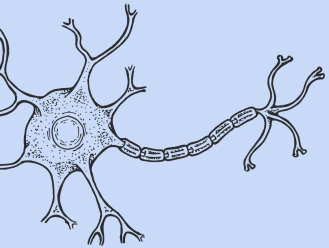
Opis problema

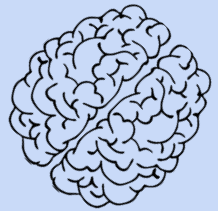
Opis rješenja

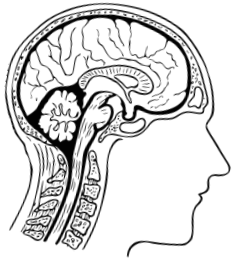
Usporedba modela

Rezultati i problemi

Zaključak





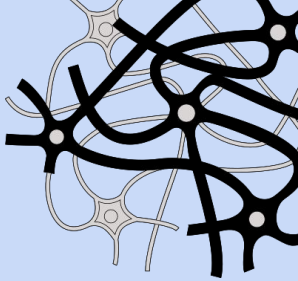


OPIS PROBLEMA

- **CT mozga** je brza i široko dostupna metoda snimanja u hitnoj medicini
- Omogućuje vizualizaciju **struktura mozga i akutnih promjena**
- **Intracerebralno krvarenje** predstavlja hitno, životno ugrožavajuće stanje
- Na CT slikama se očituje kao **hiperdenzne (svjetlije) regije**
- Rano prepoznavanje krvarenja ključno je za **pravovremeno liječenje**
- Ručna analiza CT snimaka može biti **vremenski zahtjevna i subjektivna**



Dataset i analiza podataka



- **Izvor podataka:** Kaggle – *Head CT Hemorrhage*
- **Tip podataka:** 2D CT slike mozga (PNG/JPG) & CSV datoteka s oznakama (krvarenje / bez krvarenja)
- **Struktura skupa podataka:**
  Jedna slika = jedan CT presjek
- **Raspodjela klasa:** 100 slika krvarenja i 100 bez krvarenja **= ravnoteža**
- **Uočeno:** Krvarenja se pojavljuju kao svjetlija (hiperdenzna) područja

Formulacija problema:

- Binarna klasifikacija:
  - krvarenje
  - bez krvarenja

```
id, hemorrhage
0,1
1,1
2,1
3,1
4,1
5,1
6,1
7,1
8,1
9,1
10,1
11,1
12,1
13,1
14,1
15,1
16,1
```

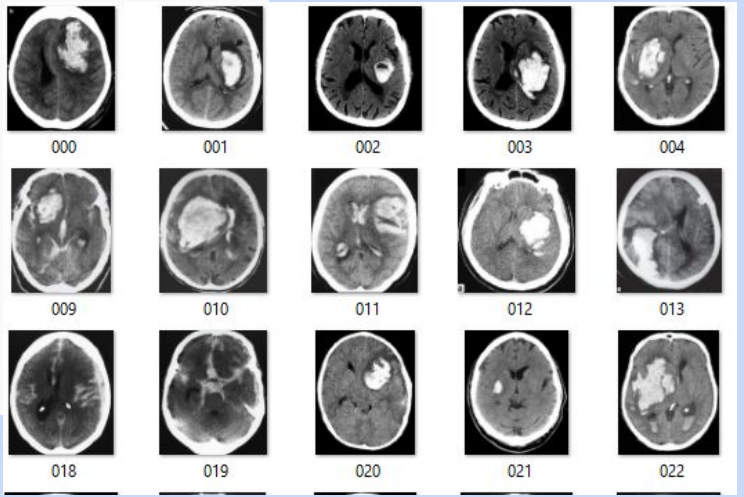
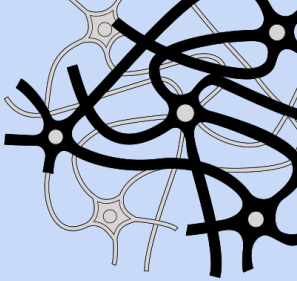




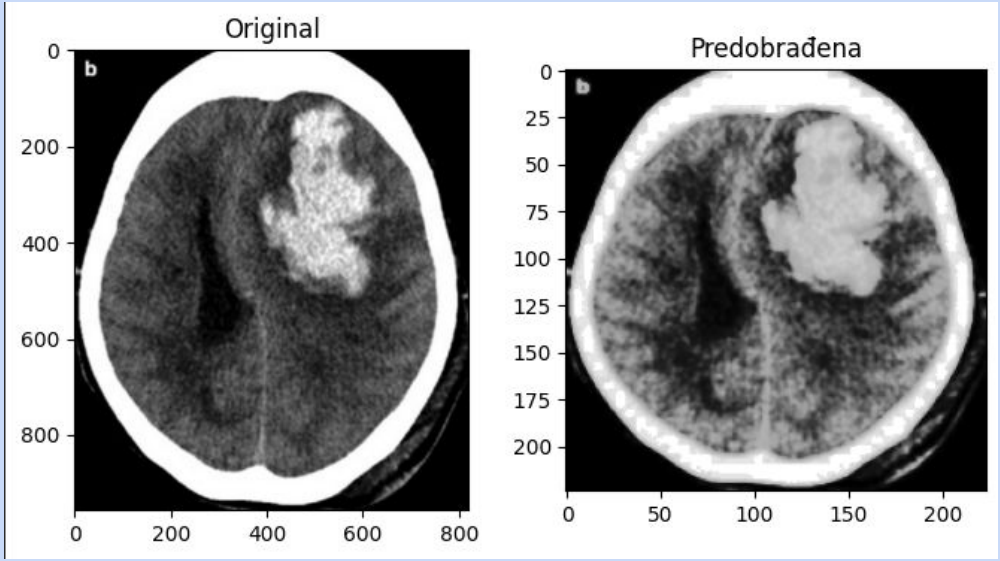
Image Preprocessing



Normalizacija slike – sve CT slike skalirane na 224×224 px i pretvorene u grayscale

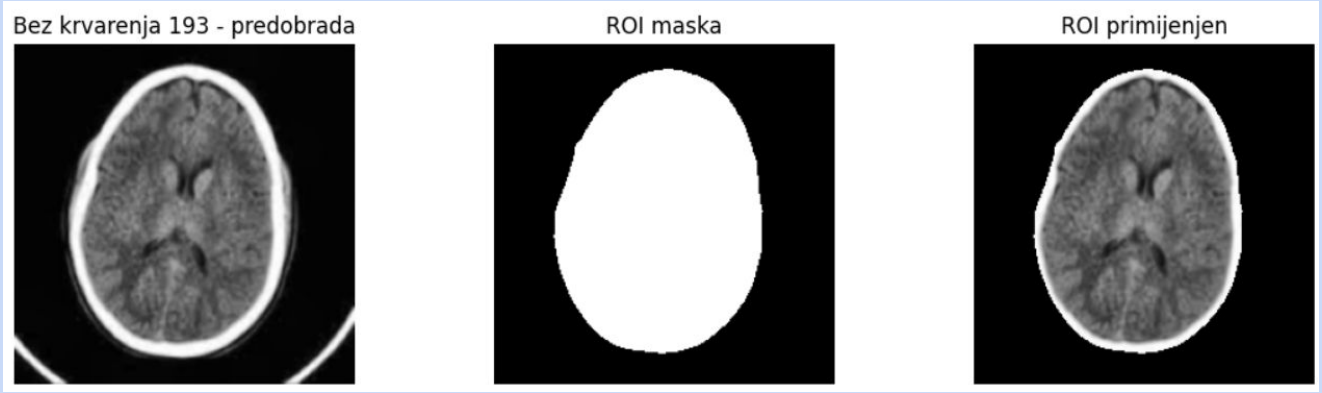
Uklanjanje šuma – Gaussian filtriranje za smanjenje artefakata

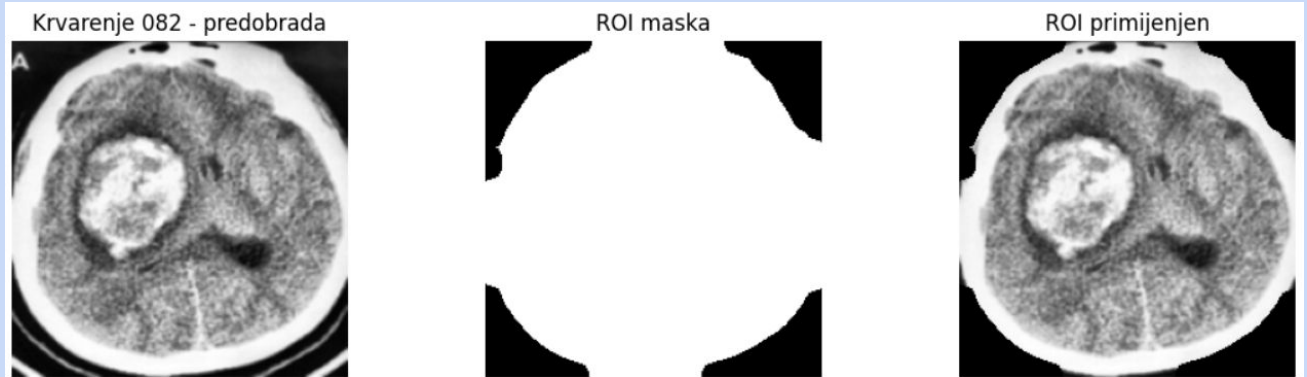
Poboljšanje kontrasta – CLAHE metoda za jasnije isticanje krvarenja



Segmentacija

ROI (Region of Interest) predstavlja **područje mozga unutar lubanje**, bez pozadine i koštanih struktura, teksta itd.





Thresholding visokih intenziteta
→ izdvajamo kost lubanje

Morfološko zatvaranje
→ spajamo diskontinuirane dijelove prstena lubanje

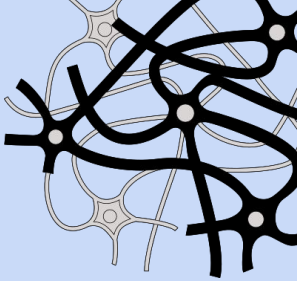
Najveća povezana komponenta
→ zadržavamo samo lubanju

Popunjavanje unutrašnjosti (binary fill holes)
→ dobivamo cijeli mozak kao ROI

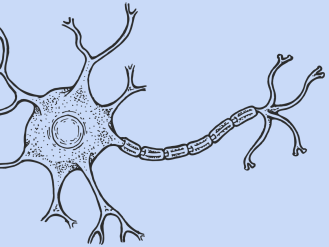
Erozija
→ uklanjamo rubove kosti iz ROI-ja

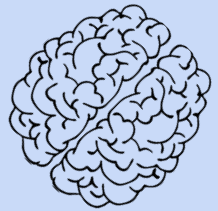


Izdvajanje značajki (FEATURE EXTRACTION)



- izdvajanje značajki koje kvantitativno opisuju sadržaj CT slike
- pretvoriti vizualne informacije iz slike u numeričke deskriptore
- tri komplementarne skupine značajki: intenzitetske, rubne i teksturalne značajke



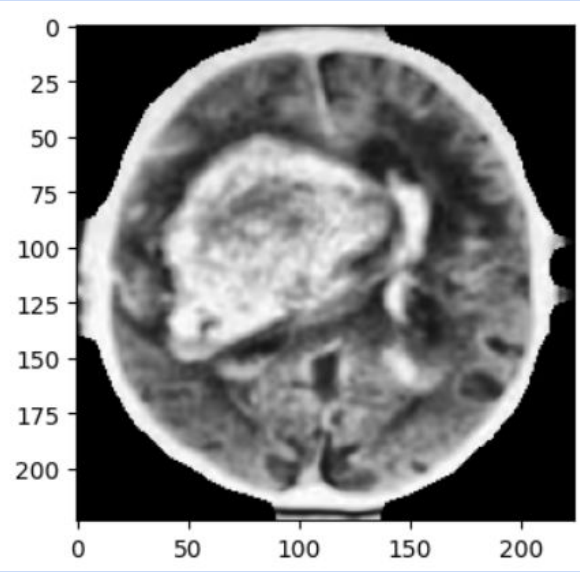


Intenzitetske značajke

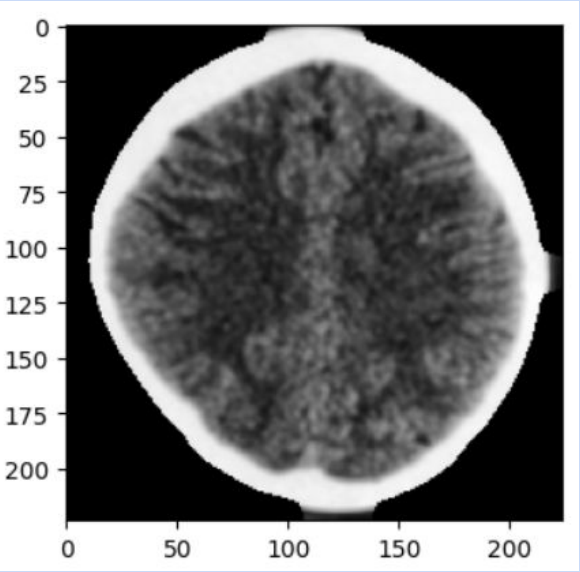


Krvarenje je "hiperdenzno", to znači da je krv gušća od okolnog mozga i zato "svijetli" na slici

1.



2.




```
1.  Features for Hemorrhage Image
Mean: 140.71
Std:  63.23
Skew: 0.15
Kurt: -1.14
Bright %: 0.3711

2. Features for No Hemorrhage Image
Mean: 112.41
Std:  70.80
Skew: 0.88
Kurt: -0.70
Bright %: 0.2385
```

Značajke temeljene na rubovima



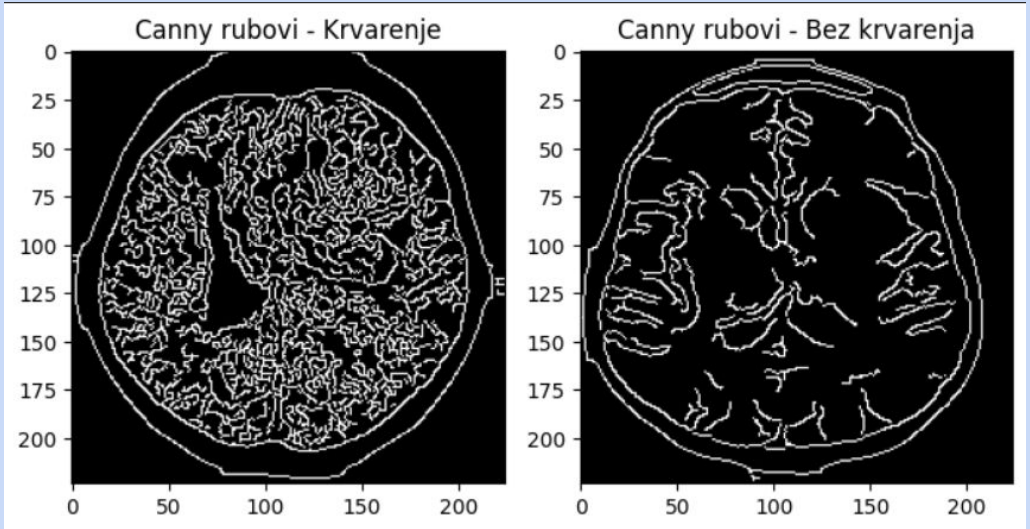
Detekcija rubova (npr. Canny ili Sobel) kako bi se istaknule nagle promjene intenziteta.

Gustoća rubova (edge density) – omjer broja rubnih piksela i ukupnog broja piksela u ROI-u

Prosječna jačina gradijenta – mjeri koliko su prijelazi intenziteta izraženi

1.

2.




```
1.  Features for Hemorrhage Image
Avg Gradient: 117.89
Edge Density: 0.2173

2. Features for No Hemorrhage Image
Avg Gradient: 101.63
Edge Density: 0.1063
```

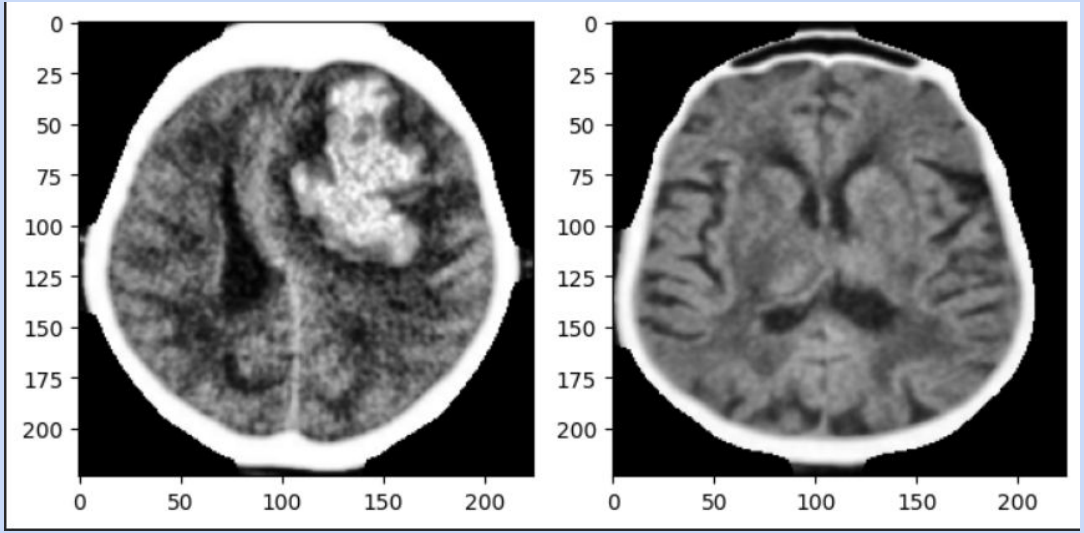
Teksturalne značajke



- LBP (Local Binary Patterns) - mikrostruktura slike, nepravilnosti i granularnost teksture
- GLCM (Gray Level Co-occurrence Matrix) - moguće je izdvojiti značajke poput: kontrasta, homogenosti, energije, korelacije

1.

2.




```
1.  Features for Hemorrhage Image
Contrast:    623.27
Homogeneity: 0.43
Energy:      0.28
Correlation: 0.96
LBP Mean:    4.76
LBP Std:     2.10

2. Features for No Hemorrhage Image
Contrast:    508.39
Homogeneity: 0.44
Energy:      0.31
Correlation: 0.95
LBP Mean:    4.44
LBP Std:     1.77
```

Treniranje modela

Ulazni podatak: **13 izdvojenih značajki po slici**

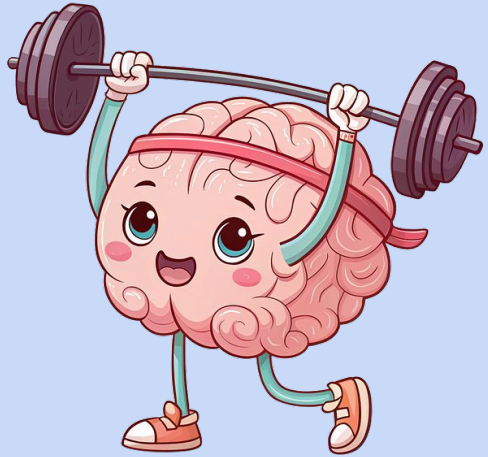
Značajke **standardizirane** prije treniranja modela

Podjela podataka: **80% trening - 20% test**

Korišteni modeli:

- logistička regresija
- SVM
- Random Forest
- XGBoost
- MLP (neuronska mreža)

Cilj: **binarna klasifikacija** (krvarenje / bez krvarenja)



Evaluacija modela

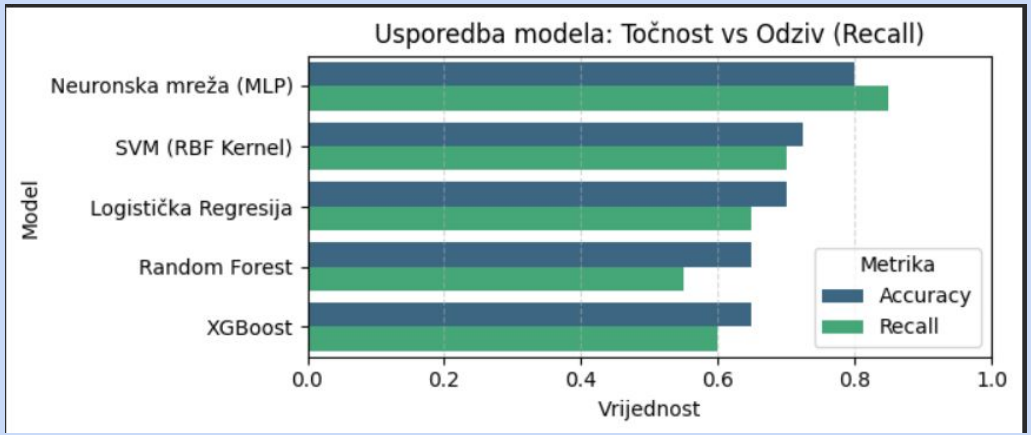
Točnost/*Accuracy*

Odziv/*Recall*:

- *Koliki udio stvarnih krvarenja je model ispravno detektirao*

U medicinskom kontekstu:

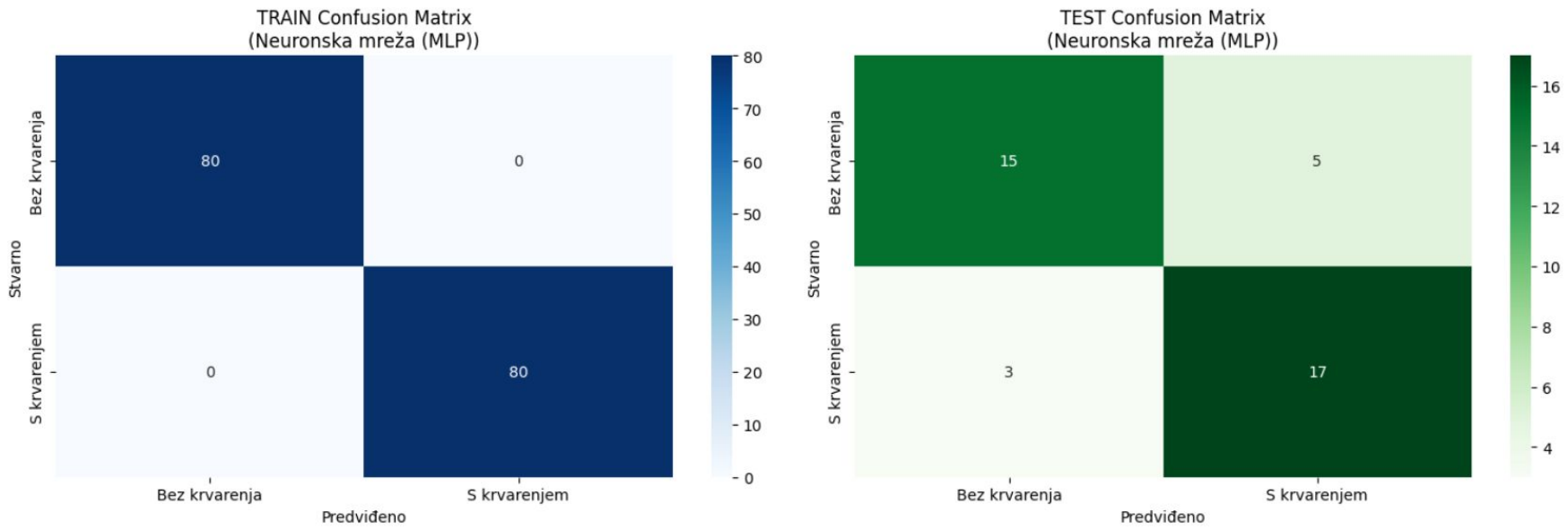
- **važnije je NE PROPUSTITI krvarenja**
- prihvatljiva je nešto manja točnost ako je **recall visok**

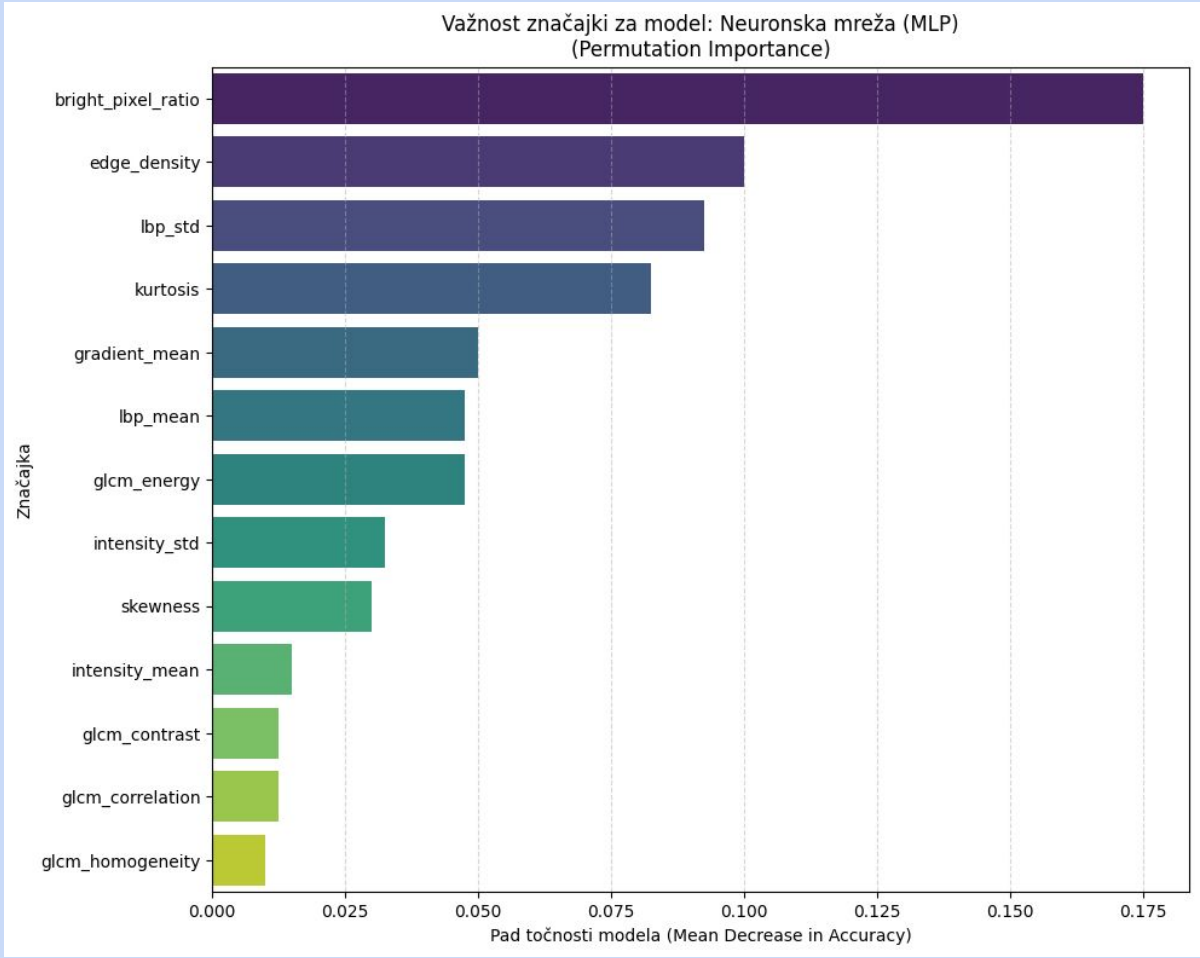


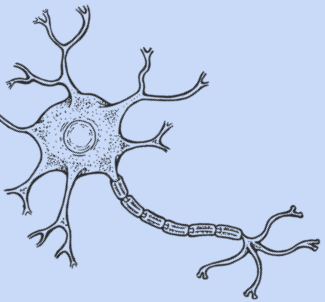
Rezultati

Model	Accuracy - Train	Recall - Train	Accuracy - Test	Recall - Test
Logistička regresija	71.88%	0.70	70.00%	0.65
SVM	90.62%	0.83	72.50%	0.70
Random Forest	99.38%	**1.00**	65.00%	0.55
XGBoost	**100%**	**1.00**	65.00%	0.60
MLP	**100%**	**1.00**	**80.00%**	**0.85**

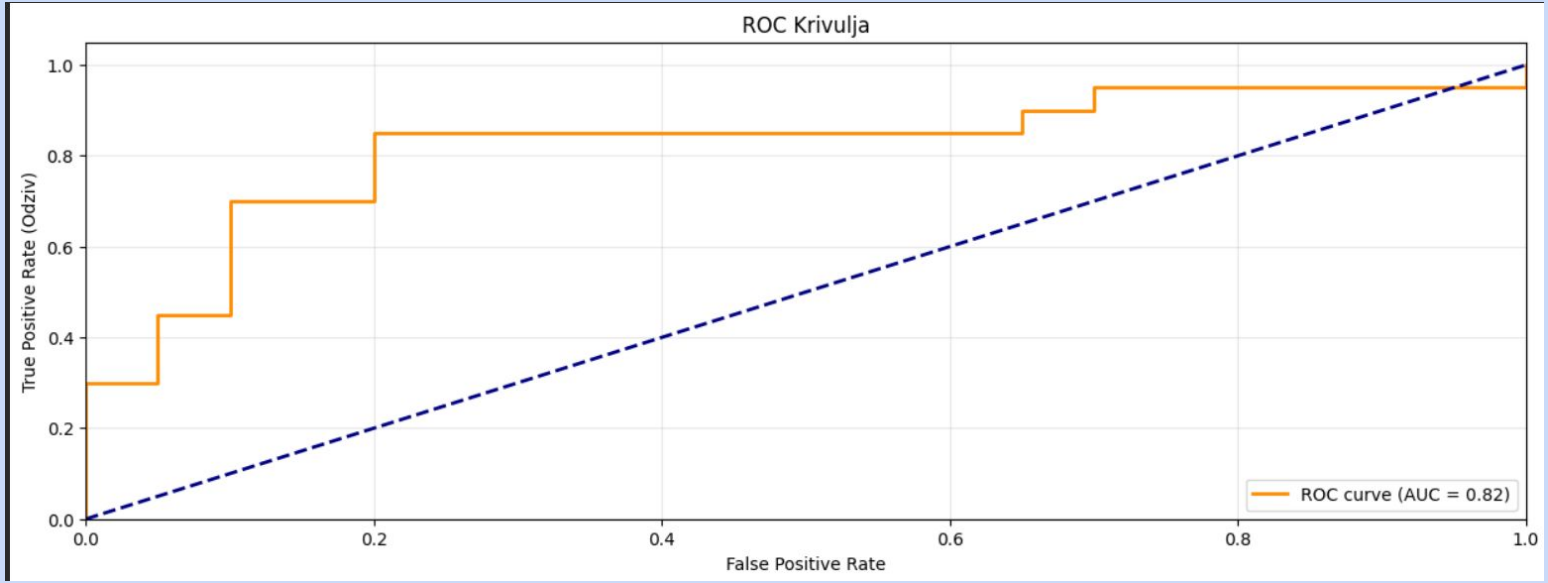
Najbolji model - matrice zabune




Važnost značajki za model: Neuronska mreža (MLP)
(Permutation Importance)
bright_pixel_ratio
edge_density
lbp_std
kurtosis
gradient_mean
lbp_mean
glcm_energy
intensity_std
skewness
intensity_mean
glcm_contrast
glcm_correlation
glcm_homogeneity
Značajka
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
Pad točnosti modela (Mean Decrease in Accuracy)



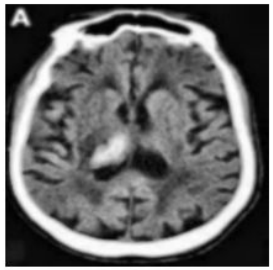



ROC Krivulja
True Positive Rate (Odziv)
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
ROC curve (AUC = 0.82)

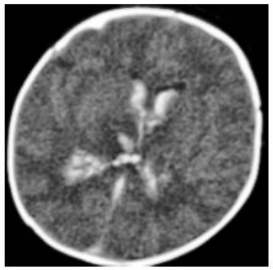
Greške modela - Poboljšanja?

FALSE NEGATIVE: Ima krvarenje -> Model kaže zdrav

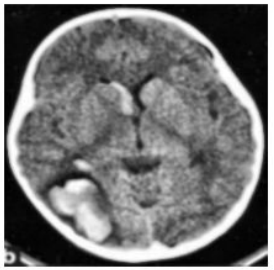
ID: 6



ID: 72

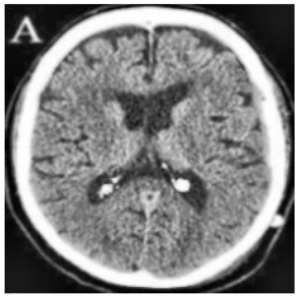


ID: 32

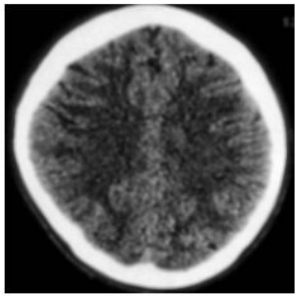


FALSE POSITIVE: Zdrav -> Model kaže krvarenje

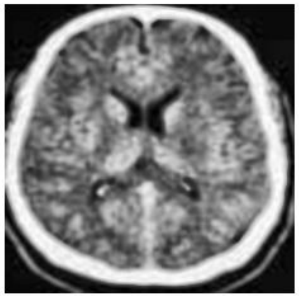
ID: 114



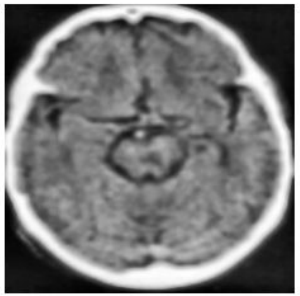
ID: 171



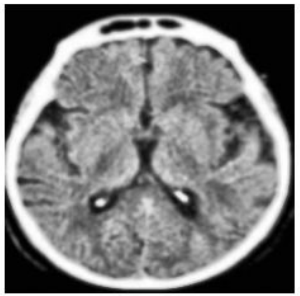
ID: 120



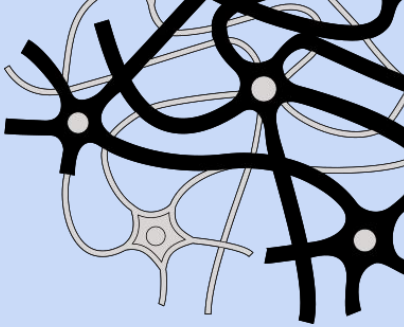
ID: 102



ID: 151







HVALA

